

Experts' views

"Historically, one of the biggest barriers to automation and robotic adoption has been cost and time constraints associated with the production setup. Traditionally, setup is done by a team of robotics experts adept in programming languages and advanced engineering. The rise in collaborative applications has made robots easily deployable without the need for companies to have large and complex engineering departments. These offer a fast, easy, and affordable alternative to bulky traditional industrial robots. Intelligent robotic tooling comes with in-built features that are easily programmable, enabling faster set-up. These solutions are making automation more inclusive, enabling collaborative applications to meet modern automation needs."

—Niels Ole Sinkbæk Sørensen, country manager, South Asia, Southeast Asia & Oceania, OnRobot

"The disadvantage is that mobile professional service robotics works on batteries, and power often restricts their functioning accomplishments. Also, running on-the-go machine learning algorithms using traditional chips such as GPUs that utilise hundreds or even thousands of watts leads to a battery capacity limitation. Last but not the least, maintaining strong and reliable connectivity in professional robotics for their constant mobility is often difficult. And using 4G networks connectivity or inexpensive Wi-Fi systems in robotics do not guarantee their quality of service. We can make use of advances in edge AI chips, which can improve the robotics performance and power consumption as these chips are designed to draw much lesser power than GPU chips. Advanced 5G technology in solving connectivity problems gives a reliability rate of 99.99 per cent. Also, network slicing technology allocates network performance to different tasks based on their priority with enhanced reliability. 5G delivers sub-millisecond latencies, which allow much faster reaction times."

—Sangeet Kumar, CEO and founder, Addverb

"I would say that the benefits outweigh the challenges. The industry—tier 1 and 2, OEMs and suppliers, large and small—realise the fact that this is not yet another passing tech fad. Everyone is working towards technological transformation. That said, this change is effort and capital intense. The scale, pace, and levels of automation are different across organisations. To top it all, there is the pressure to ensure that your workforce is skilled and fully equipped to manage the change. There are other aspects too to consider, for example, making sense of the data we generate and convert it into usable information. Another example is cybersecurity."

—Phanindra Karody, head, Bangalore Plant, Continental Automotive India

possible time, the businesses have been quick to adopt robots and cobots on the shop floor.

Karody further adds, "These collaborative robots help in relieving staff from the repetitive and mundane tasks and eliminate the inaccuracies caused due to fatigue and manual errors. Businesses can look for the sweet spot by carefully automating the menial and physically taxing tasks and delegating the more adroit and skilful tasks to humans. This sweet spot can ensure highly scalable and accurate operations. With the improvement in technology and

IT infrastructure, the application of robots and cobots can see a disruptive improvement and lead to far wider adoption by businesses."

A report in the International Federation of Robotics (IFR) suggests that a substantial share of this cobot adoption will be driven by more intelligent components such as smart grippers, plug & play interfaces, and programming by demonstrations. Programming by demonstration allows direct programming methods through digital sensors and smart software. For instance, workers can now program a robot by taking the robot arm and

hand guiding it through the movements.

Niels Ole Sinkbæk Sørensen, country manager, South Asia, South-east Asia & Oceania, OnRobot, says, "Robots offer numerous benefits, including making the production process shorter, increasing productivity, reducing the defect rate and supporting employees, making them more efficient and valuable. Automation is crucial, especially for SMEs who have smaller cash reserves and a more limited customer base, making them vulnerable to disruptions. These companies have new options in automation as collaborative applications offer a fast, easy, and affordable alternative to bulky traditional industrial robots. Collaborative robots (cobots)—robots that work safely alongside workers—account for three percent of global robot sales, and this share is expected to rise to 34 per cent of a 25 billion dollars market by 2025."

Growth of robotics market in India and globally

According to IFR, Asia is said to be the largest market for robotics. Back in 2018, robotics installations in South Korea and China chiefly reduced, while the market for Japan augmented massively. Overall, the market in Asia grew by one per cent. Both Europe and the US reached the peak of robotics installations six years in a row.

The new World Robotics report shows an annual global sales value of 16.5 billion dollars in 2018—a new record. 422,000 units were shipped globally in 2018—an increase of six per cent compared to the previous year. IFR predicts an average growth of twelve percent per year from 2020 to 2022. Allied Market Research forecasts that the industrial robotics market size is expected to reach 70,715 million dollars by 2023, at a CAGR of 9.4 per cent from 2017 to 2023. Statista, on the other hand, claims that the global market for robots is expected to grow at a CAGR of around 26 per cent to reach 210 billion dollars by 2025. It is predicted that this market will hit the hundred billion dollars mark in 2020.

Talking about the Indian market,

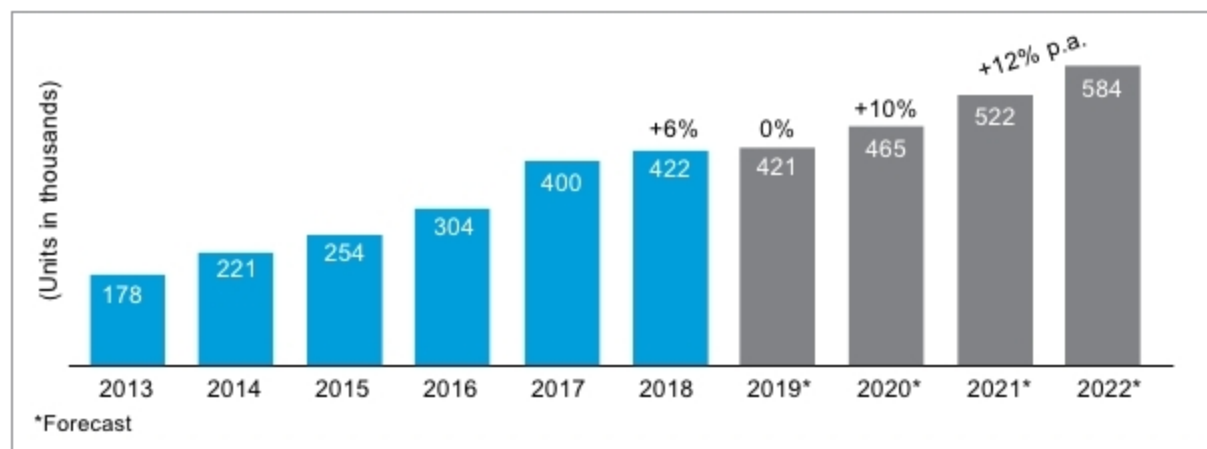


Fig. 2: Annual installations of industrial robots 2013-2018 and 2019*-2022* (Source: World Robotics 2019)